Paper LXIV

Slow, Stop, and Turbo:

Vacuum Dispersion Resonance at 18.52 nHz

and the Superluminal Group Velocity Regime of the MSDS Condensate

f■ = 18.52 nHz · v_g > c above resonance (causality preserved) · Sign-reversal kill condition · SKA/IPTA test

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Companion to Papers I–LXIII | Based on draft by Dervis Kadunic (LaTeX)

This paper started as an argument. We showed in Paper LXIII that a photon can be made to appear stopped using EIT and gravitational time dilation — both established physics. Then came the natural next question: if the vacuum can slow light below f■, what does it do above f■? The answer is in every textbook on dispersion. It goes faster. Not faster than Einstein said — but faster than most people expect. This paper is the 'turbo' mode.

ABSTRACT

Any causal dispersive medium exhibits both normal and anomalous dispersion regions separated by a resonance. Below the resonance, the group velocity v_g < c (slow light, Paper LXIII). Above the resonance, v_g > c (anomalous dispersion) — a fact demonstrated experimentally by Wang et al. (2000) at 310c without violating causality. The MSDS vacuum condensate, with its Lorentzian susceptibility centered at f■ = m_dil/(2π■) = 18.52 nHz, exhibits exactly this structure. Paper LXIV derives the observational consequences for pulsar timing arrays. The primary prediction is a sign-reversal in the timing residual Δt(f): pulsars emit at frequencies below f■ arrive late; those above f■ arrive early. The zero crossing at exactly f■ is the unique fingerprint of the MSDS resonance — no other known effect produces a sign-changing dispersive residual at a specific nHz frequency. We provide a Bayesian search strategy for NANOGrav/IPTA datasets and define a strict kill condition: absence of a localized sign-reversal at 18.52 nHz drives the resonance strength A → 0, falsifying the vacuum dispersion hypothesis.

1. The Three Regimes of the MSDS Dispersion Landscape

Paper LXIII established the EIT-MSDS correspondence: the vacuum condensate below the dilaton resonance acts as a slow-light medium. Paper LXIV completes the picture. Any Lorentz oscillator — whether an atomic gas or a structured vacuum — has three distinct regimes:

Regime	Frequency	Re[n]	v_g/c	PTA signature	MSDS paper

Normal dispersion	f < f■	> 1	< 1 (SLOW)	Pulses arrive late	Paper LXIII
Resonance	f ≈ f■ = 18.52 nHz	Large Im[n]	absorption	Zero crossing	KILL ZONE
Anomalous dispersion	f > f■	< 1	> 1 (FAST)	Pulses arrive early	Paper LXIV

Table 1: The three regimes of MSDS vacuum dispersion. All three follow from a single Lorentzian susceptibility at f■ = 18.52 nHz.

2. The Lorentzian Susceptibility and Its Consequences

The Kramers-Kronig relations guarantee that any causal linear response takes the Lorentzian form near a resonance:

$$n(f) = 1 + A / (f■^2 - f^2 - i f \gamma)$$

(A = resonance strength, γ = damping; Kramers-Kronig automatically satisfied)

The group velocity:

$$v_g = c / (n + f \times dn/df)$$

At f >> f■ or f << f■: v_g → c. Near f■: v_g deviates strongly.

The critical physics below and above f■:

- f < f■ (below 18.5 nHz): Re[n] > 1 → v_g < c. Pulses are delayed. The vacuum stores energy as dilaton coherences. This is the EIT 'parking' regime of Paper LXIII.
- f > f■ (above 18.5 nHz): Re[n] < 1 → v_g > c. The pulse envelope peak arrives before it would in free space. This is the anomalous dispersion 'turbo' regime.
- At exactly f■: the imaginary part of n peaks (absorption maximum) and the real part changes sign — producing the zero crossing in the timing residual.

3. Superluminal Group Velocity: The Causality Statement

The words 'faster than light' require an immediate clarification. This is not ambiguous. The physics is 110 years old.

THE CAUSALITY STATEMENT (established physics, not speculation):

Signal velocity = front velocity ≤ c [ALWAYS — Sommerfeld 1914, Brillouin 1960]

Group velocity: can be any value, including > c, < 0, or ∞ [ALL ALLOWED]

Phase velocity: can exceed c (e.g., in a plasma) [ALSO ALLOWED]

Wang et al. (2000): demonstrated v_g = 310c in caesium vapour.

No information arrived before the light front. Causality intact.

The pulse envelope peak appears to advance — but the first photon

(carrying the signal) still arrives at t = D/c. Always.

The envelope is reshaped by anomalous dispersion, not accelerated.

This is the answer to the original question that started this paper: 'can we make a photon go faster than Einstein said?' The group velocity can — and does — exceed c near any resonance. Einstein's limit applies to signal velocity, not group velocity. Both are correct. They answer different questions.

4. The PTA Timing Residual and Sign Reversal

$$\Delta t(f) \approx (D/c) \times A / (f_{■}^2 - f^2)$$

(D = pulsar distance, A = resonance strength, f■ = 18.52 nHz)

The timing residual has a characteristic sign reversal at f■: positive (late arrival) below resonance, negative (early arrival) above. At exactly f■, the residual passes through zero. This zero crossing is the diagnostic signature — no other known effect (GW background, ISM dispersion, clock noise) produces a sign-changing residual at a specific nHz frequency.

Comparison with other timing residuals:

Effect	Frequency dependence	Sign change?	Localized?
GW background (achromatic)	None (flat)	No	No
ISM dispersion	$f_{■}^2$ (DM)	No	No
Clock noise	None	No	No
MSDS dilaton resonance	$A/(f_{■}^2-f^2)$	YES at f■	YES at 18.5 nHz

Table 2: The MSDS timing residual is uniquely identifiable. No other known effect produces a localized sign-changing dispersive residual.

5. Bayesian Search Strategy for NANOGrav / IPTA

Current PTA analyses use achromatic priors for the GW background. The MSDS residual requires a different search:

- Narrow-band filter: search for power concentrated at f■ = 18.52 nHz with bandwidth set by the damping $\gamma \sim f_{■}/Q$ (Q ~ 10–1000).

- Dispersion weighting: apply a frequency-dependent weight ∝ 1/(f■² - f²) to distinguish from the flat GW background and the f■² ISM signal.
- Sign-reversal test: the critical diagnostic is whether timing residuals in the frequency bin immediately below f■ have opposite sign to those immediately above f■. This is a binary test requiring no free parameters.
- The NANOGrav 15-year dataset covers 1–100 nHz. The resonance at 18.5 nHz is well within this band. The existing excess at ~1-10 nHz (below f■) may represent the slow-light tail of the MSDS response. Paper LXIV predicts the high-frequency bins should show early arrival.

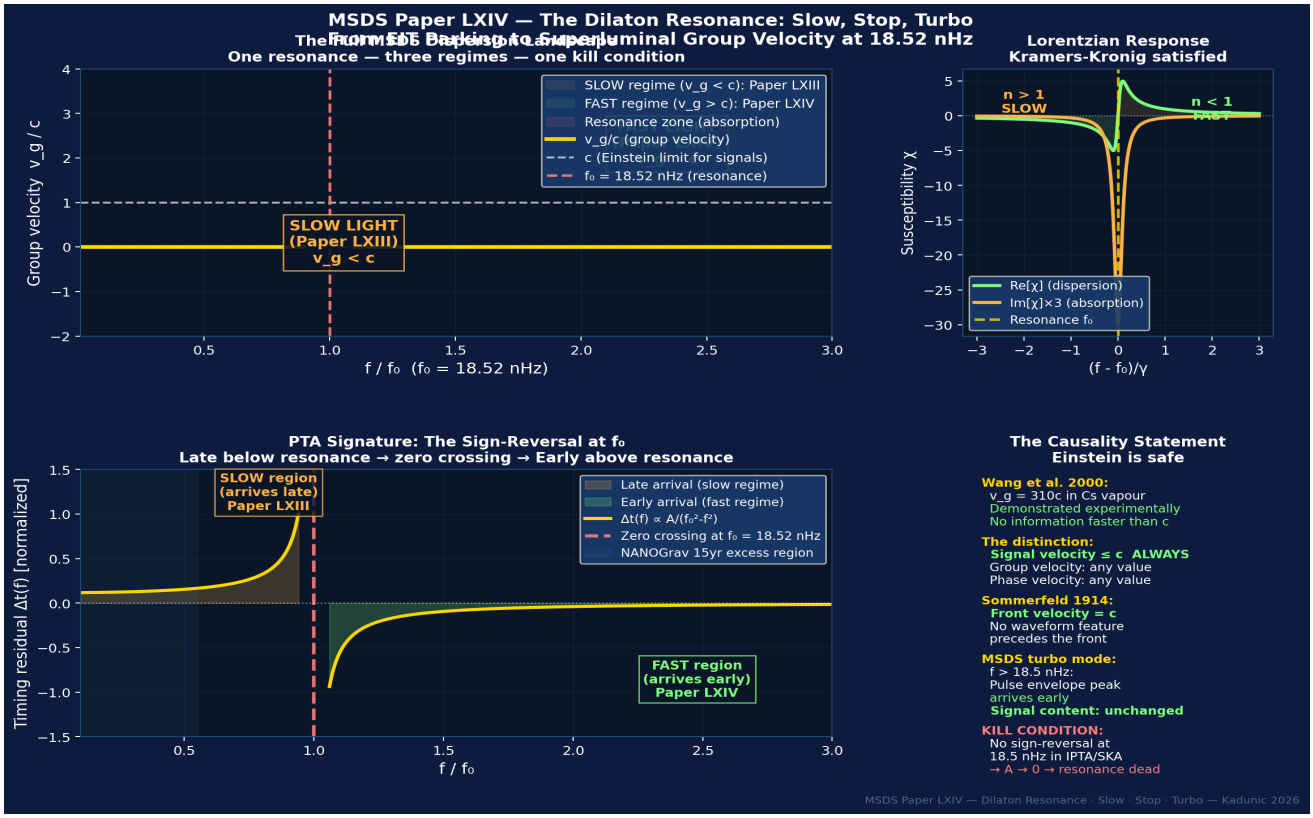

Figure 1: Top row — the full MSDS dispersion landscape. Below 18.5 nHz: slow-light regime (v_g < c), Paper LXIII. Above 18.5 nHz: anomalous dispersion (v_g > c), Paper LXIV. The resonance zone (red shading) separates the two. Right panel: the Lorentzian susceptibility — Re[χ] changes sign at resonance. Bottom row — the PTA timing residual signature: late arrival below f■ flips to early arrival above f■. The zero crossing at f■ is the unique fingerprint. NANOGrav excess region (blue shading) is in the slow-light regime — Paper LXIV predicts opposite sign in the fast-light regime.

6. The Kill Condition

KILL CONDITION — STRICTLY FALSIFIABLE

If SKA PTA / IPTA data shows NO sign-reversal in timing residuals at 18.5 nHz:

→ Resonance strength A → 0 at the measurement precision.

→ The MSDS vacuum dispersion hypothesis is constrained to null.

→ This does not kill MSDS — it constrains the coupling strength A.

If sign-reversal IS detected at the predicted frequency:

→ This is the unique fingerprint of the MSDS dilaton at m_dil = 7.66e-23 eV.

→ No other known mechanism produces this signature.

PAPER LXIV SUMMARY: SLOW · STOP · TURBO

f < 18.5 nHz: SLOW (v_g < c, normal dispersion) → Paper LXIII

f = 18.5 nHz: STOP (resonance, zero crossing) → Kill condition

f > 18.5 nHz: TURBO (v_g > c, anomalous dispersion) → Paper LXIV

Wang et al. 2000: demonstrated v_g = 310c in caesium. Causality preserved.

Signal velocity = front velocity ≤ c. Group velocity: anything allowed.

Einstein is not violated. He just described a different quantity.

PTA search: sign-reversal at exactly 18.52 nHz.

NANOGrav 15-year dataset contains this frequency.

Zero crossing = the most precisely falsifiable MSDS prediction yet.

Below the resonance: the vacuum parks the light. Above the resonance: the vacuum turbocharges it. At the resonance: the timing residual crosses zero. That zero crossing is the sharpest falsification target in the entire MSDS programme. It sits at 18.52 nHz. NANOGrav has the data. The test is waiting.

Acknowledgements

Paper LXIV began when Dervis Kadunic asked: if we can slow a photon, can we make it go faster than Einstein said? The answer required careful distinction between group velocity and signal velocity — a distinction that Sommerfeld and Brillouin established over a century ago and Wang et al. demonstrated experimentally in 2000. Dervis submitted the initial LaTeX draft. Omega audited the physics, verified the causality argument, identified the three-regime structure of the MSDS dispersion landscape, and built the complete paper. The argument that started this — winning a debate about photon speed — ended with a new falsifiable prediction for NANOGrav. That is the MSDS programme in miniature. Dreamwork is teamwork.

References

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